Total Factor Productivity (TFP): Aggregation, Misallocation, and Dynamic Decomposition

Coherent Revision Notes with Full Derivations and Proofs

1 Why aggregation and decomposition are non-trivial

A central object in growth and productivity analysis is *aggregate* total factor productivity (TFP), typically defined through an aggregate production function such as

$$Y = A\,K^{\alpha}L^{1-\alpha}, \qquad \alpha \in (0,1), \tag{1}$$

where Y is aggregate output, K and L are aggregate capital and labor, and A is aggregate TFP.

Key message. Aggregate TFP is not a simple arithmetic average (or sum) of firm-level efficiencies. Even if each firm has a well-defined technology parameter, the mapping from micro efficiency to aggregate A depends on *how inputs are allocated* across heterogeneous producers. Misallocation (due to wedges, frictions, regulations, or imperfect markets) breaks the equivalence between "better micro technology" and "higher aggregate TFP".

These notes build a single logical chain:

1. Start with micro-to-macro aggregation under heterogeneity and wedges (the Hsieh–Klenow style logic).
2. Extract a measurable misallocation statistic (dispersion of revenue productivity).
3. Move to empirical, *accounting-style* decompositions of aggregate productivity changes (BHC/GR/FHK/OP and dynamic OP variants).
4. Extend the accounting logic to sectoral reallocation and investment allocation (APPF and Wurgler-style elasticities).

2 Micro-to-macro aggregation with misallocation (Hsieh–Klenow logic)

2.1 Three-layer structure: final good, sector, firm

Assumption 2.1 (Final good aggregator)**.** There are S sectors indexed by $s = 1, \ldots, S$. The final good is produced competitively using a Cobb–Douglas aggregator:

$$Y = \prod_{s=1}^{S} Y_s^{\theta_s}, \qquad \theta_s > 0, \quad \sum_{s=1}^{S} \theta_s = 1. \tag{2}$$

The final good price is normalized to one.

Proposition 2.1 (Final-good cost minimization implies sector demand index)**.** *Let P_s be the sector-s price index. Under perfect competition in the final good market, the first-order condition implies*

$$P_s = \theta_s \frac{Y}{Y_s}. \tag{3}$$

Proof. A competitive final-good producer chooses $\{Y_s\}$ to maximize $Y - \sum_s P_s Y_s$ with Y given by (2). The derivative is

$$\frac{\partial Y}{\partial Y_s} = \theta_s \frac{Y}{Y_s}.$$

Setting $\partial/\partial Y_s$ of the objective to zero yields $\theta_s Y/Y_s = P_s$, establishing (3). $\square$

Assumption 2.2 (Sector aggregator and firm demand)**.** Within each sector s, there are M_s differentiated varieties (firms) indexed by $i = 1, \ldots, M_s$. Sector output is CES:

$$Y_s = \left(\sum_{i=1}^{M_s} y_{si}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \qquad \sigma > 1, \tag{4}$$

with the associated demand system

$$y_{si} = Y_s \left(\frac{p_{si}}{P_s} \right)^{-\sigma}, \qquad P_s = \left(\sum_{i=1}^{M_s} p_{si}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}. \tag{5}$$

Proposition 2.2 (Markup pricing under CES demand)**.** *For any firm i in sector s, revenue $p_{si} y_{si}$ satisfies*

$$p_{si} = \frac{\sigma}{\sigma - 1} MC_{si} \cdot \frac{1}{1 - \tau_{si}^Y}, \tag{6}$$

where MC_{si} is marginal cost and τ_{si}^Y is an output wedge (a tax if positive, a subsidy if negative) that scales effective revenue.

Proof. Under CES demand, the elasticity of demand is σ, so a monopolistic competitor sets a constant markup $\sigma/(\sigma - 1)$ over marginal cost. If an output wedge reduces effective revenue by factor $(1 - \tau_{si}^Y)$, the firm acts as if its effective price is $(1 - \tau_{si}^Y)p_{si}$, so the pricing rule becomes $(1 - \tau_{si}^Y)p_{si} = \frac{\sigma}{\sigma-1} MC_{si}$, which is equivalent to (6). $\square$

2.2 Firm technology and wedges

Assumption 2.3 (Firm production and wedges)**.** Each firm has Cobb–Douglas production:

$$y_{si} = A_{si}\, k_{si}^{\alpha_s}\, \ell_{si}^{1-\alpha_s}, \qquad \alpha_s \in (0, 1), \tag{7}$$

with wage w and rental rate R. Firms face:

- an *output wedge* τ_{si}^Y multiplying revenue by $(1 - \tau_{si}^Y)$;
- a *capital wedge* τ_{si}^K multiplying the effective unit cost of capital by $(1 + \tau_{si}^K)$.

Profits are

$$\pi_{si} = (1 - \tau_{si}^Y) p_{si} y_{si} - w\ell_{si} - (1 + \tau_{si}^K) R k_{si}. \tag{8}$$

2.3 TFPQ vs TFPR and why TFPR dispersion measures misallocation

Definition 2.1 (Physical productivity (TFPQ))**.** A_{si} in (7) is *physical* productivity (quantity-based productivity).

Definition 2.2 (Revenue productivity (TFPR))**.** Define revenue productivity as

$$\text{TFPR}_{si} := \frac{p_{si} y_{si}}{k_{si}^{\alpha_s} (w\ell_{si})^{1-\alpha_s}}. \tag{9}$$

Proposition 2.3 (No wedges $\Rightarrow$ TFPR equalization within a sector). *If $\tau^Y_{si} = 0$ and $\tau^K_{si} = 0$ for all i in a fixed sector s, then in an interior optimum all active firms in sector s share a common* TFPR_{si}.

Proof. With no wedges, (8) implies first-order conditions (FOCs)

$$p_{si}\frac{\partial y_{si}}{\partial \ell_{si}} = w, \qquad p_{si}\frac{\partial y_{si}}{\partial k_{si}} = R.$$

Using Cobb–Douglas (7),

$$\frac{\partial y_{si}}{\partial \ell_{si}} = (1-\alpha_s)\frac{y_{si}}{\ell_{si}}, \qquad \frac{\partial y_{si}}{\partial k_{si}} = \alpha_s\frac{y_{si}}{k_{si}}.$$

Thus

$$(1-\alpha_s)\frac{p_{si}y_{si}}{\ell_{si}} = w, \qquad \alpha_s\frac{p_{si}y_{si}}{k_{si}} = R.$$

Rearrange the two equalities:

$$p_{si}y_{si} = \frac{w}{1-\alpha_s}\,\ell_{si} = \frac{R}{\alpha_s}\,k_{si}.$$

Combining them yields the common capital-labor ratio $k_{si}/\ell_{si} = \frac{\alpha_s}{1-\alpha_s}\frac{w}{R}$, which is identical across firms in sector s when prices w, R are common. Substituting $p_{si}y_{si}$ into (9) shows TFPR_{si} depends only on (w, R, α_s) and constants, hence is identical across active firms in sector s. □

Remark 2.1 (Interpretation). If TFPR_{si} differs across firms within the same sector, then the marginal revenue products of inputs differ across firms, so reallocating inputs can increase output without changing technology: this is precisely misallocation.

2.4 Efficient output vs observed output

Let Y be observed output under wedges. Let Y^{eff} be counterfactual output if wedges were removed while keeping the same sectoral aggregates and technologies. A standard representation (consistent with the Hsieh–Klenow accounting identity) is:

$$\frac{Y}{Y^{\text{eff}}} = \prod_{s=1}^{S}\left[\sum_{i=1}^{M_s}\left(\frac{A_{si}}{A_s}\cdot\frac{\mathrm{TFPR}_s}{\mathrm{TFPR}_{si}}\right)^{\sigma-1}\right]^{\frac{\theta_s}{\sigma-1}}, \tag{10}$$

where A_s and TFPR_s are sector reference levels (e.g., suitable means) used to normalize units.

What (10) says. Output losses are larger when high-A_{si} firms have high TFPR_{si} (they are "too small"), or low-A_{si} firms have low TFPR_{si} (they are "too large"). In either case, dispersion of $\log \mathrm{TFPR}_{si}$ is central.

2.5 A convenient decomposition: technical vs allocative efficiency

A compact way to organize the logic is:

$$\log A^{\text{agg}} = \underbrace{\log \mathrm{TE}}_{\text{technology (TFPQ) component}} + \underbrace{\log \mathrm{AE}}_{\text{allocation (TFPR) component}}, \tag{11}$$

where:

- TE aggregates physical productivities A_{si} in a way consistent with CES aggregation (a weighted geometric-type object, with weights tied to σ);

- AE captures the penalty from misallocation; it is decreasing in the dispersion of TFPR_{si}.

Proposition 2.4 (Variance representation under lognormality (proof sketch))**.** *If, within each sector s, the pair $(\log A_{si}, \log \mathrm{TFPR}_{si})$ is jointly normal and M_s is large, then $\log(Y/Y^{\text{eff}})$ admits an approximation of the form*

$$\log \frac{Y}{Y^{\text{eff}}} \approx -\sum_{s=1}^{S} \theta_s \cdot C_s \cdot \mathrm{Var}\big(\log \mathrm{TFPR}_{si}\big), \qquad C_s > 0, \tag{12}$$

so allocative efficiency falls as $\mathrm{Var}(\log \mathrm{TFPR}_{si})$ rises.

Proof (sketch). Take logs of (10). The key object is $\log \sum_i \exp((\sigma-1)Z_{si})$ where

$$Z_{si} := \log A_{si} - \log A_s + \log \mathrm{TFPR}_s - \log \mathrm{TFPR}_{si}.$$

Under joint normality, Z_{si} is normal. For large M_s, the log-sum-exp admits a cumulant expansion where the second cumulant (variance) enters with coefficient proportional to $(\sigma-1)^2$. Because Z_{si} loads negatively on $\log \mathrm{TFPR}_{si}$, larger variance in $\log \mathrm{TFPR}_{si}$ lowers the expression, yielding (12). A fully formal proof follows from standard lognormal moment identities and bounds for log-sum-exp. □

2.6 Two "puzzles" explained by the same identity

High-tech, low-TFP. Even if TE rises rapidly (better micro technology), aggregate TFP can stagnate if AE deteriorates due to worsening misallocation.

High-distortion, high-growth. When multiple wedges exist (e.g., capital and labor), overall misallocation losses depend not only on each wedge's dispersion but also on their *covariance*. Negative covariance across wedges can partially offset losses.

Proposition 2.5 (A generic covariance offset identity)**.** *Suppose revenue productivity can be written as*

$$\log \mathrm{TFPR}_i = c + a\, X_i + b\, Y_i$$

for wedge components (X_i, Y_i) and constants (a, b). Then

$$\mathrm{Var}(\log \mathrm{TFPR}_i) = a^2 \mathrm{Var}(X_i) + b^2 \mathrm{Var}(Y_i) + 2ab\, \mathrm{Cov}(X_i, Y_i).$$

If $ab > 0$ and $\mathrm{Cov}(X_i, Y_i) < 0$, the covariance term reduces total dispersion and hence reduces misallocation losses.

Proof. Expand $\mathrm{Var}(aX + bY)$ using bilinearity of covariance:

$$\mathrm{Var}(aX + bY) = a^2 \mathrm{Var}(X) + b^2 \mathrm{Var}(Y) + 2ab\, \mathrm{Cov}(X, Y).$$

□

3 From theory to accounting: micro TFP aggregation and dynamic decomposition

3.1 Two-period setup and weighted aggregate TFP

Consider two periods $t \in \{1, 2\}$ with firm set changing over time. Let A_{it} be firm productivity and β_{it} be firm output share in period t, with $\sum_i \beta_{it} = 1$.

Definition 3.1 (Aggregate TFP as a share-weighted mean)**.**

$$A_t := \sum_{i\in\mathcal{I}_t} \beta_{it} A_{it}, \qquad \sum_{i\in\mathcal{I}_t} \beta_{it} = 1. \tag{13}$$

Partition firms:

$$S = \mathcal{I}_1 \cap \mathcal{I}_2 \quad \text{(survivors)}, \qquad X = \mathcal{I}_1 \setminus \mathcal{I}_2 \quad \text{(exiters)}, \qquad E = \mathcal{I}_2 \setminus \mathcal{I}_1 \quad \text{(entrants)}.$$

The total change is $\Delta A := A_2 - A_1$.

3.2 BHC decomposition (full derivation)

Proposition 3.1 (BHC identity)**.** *The aggregate TFP change decomposes as*

$$\Delta A = \underbrace{\sum_{i\in S} \beta_{i1}(A_{i2} - A_{i1})}_{\textit{within (productivity) effect}} + \underbrace{\sum_{i\in S} A_{i2}(\beta_{i2} - \beta_{i1})}_{\textit{between (share reallocation) effect}} + \underbrace{\sum_{i\in E} \beta_{i2} A_{i2} - \sum_{i\in X} \beta_{i1} A_{i1}}_{\textit{net entry effect}}. \tag{14}$$

Proof. Start from definitions:

$$A_2 = \sum_{i\in S} \beta_{i2} A_{i2} + \sum_{i\in E} \beta_{i2} A_{i2}, \qquad A_1 = \sum_{i\in S} \beta_{i1} A_{i1} + \sum_{i\in X} \beta_{i1} A_{i1}.$$

Thus

$$\Delta A = \left(\sum_{i\in S} \beta_{i2} A_{i2} - \sum_{i\in S} \beta_{i1} A_{i1} \right) + \sum_{i\in E} \beta_{i2} A_{i2} - \sum_{i\in X} \beta_{i1} A_{i1}.$$

Now add and subtract $\sum_{i\in S} \beta_{i1} A_{i2}$ inside the survivor bracket:

$$\begin{aligned} \sum_{i\in S} \beta_{i2} A_{i2} - \sum_{i\in S} \beta_{i1} A_{i1} &= \left(\sum_{i\in S} \beta_{i1} A_{i2} - \sum_{i\in S} \beta_{i1} A_{i1} \right) + \left(\sum_{i\in S} \beta_{i2} A_{i2} - \sum_{i\in S} \beta_{i1} A_{i2} \right) \\ &= \sum_{i\in S} \beta_{i1}(A_{i2} - A_{i1}) + \sum_{i\in S} A_{i2}(\beta_{i2} - \beta_{i1}). \end{aligned}$$

Substitute back to obtain (14). □

Remark 3.1 (Time inconsistency). In (14), the within effect uses base-period shares β_{i1}, while the between effect uses end-period productivity A_{i2}. This asymmetric choice is the "time inconsistency" concern.

3.3 GR decomposition (common reference) and its limitation

Define two-period averages for survivors:

$$\overline{\beta}_i := \frac{\beta_{i1} + \beta_{i2}}{2}, \qquad \overline{A}_i := \frac{A_{i1} + A_{i2}}{2}.$$

Proposition 3.2 (Symmetric survivor decomposition)**.** *For survivors,*

$$\sum_{i\in S} \beta_{i2} A_{i2} - \sum_{i\in S} \beta_{i1} A_{i1} = \sum_{i\in S} \overline{\beta}_i (A_{i2} - A_{i1}) + \sum_{i\in S} \overline{A}_i (\beta_{i2} - \beta_{i1}). \tag{15}$$

Proof. Use the identity

$$\beta_{i2} A_{i2} - \beta_{i1} A_{i1} = \frac{\beta_{i1} + \beta_{i2}}{2}(A_{i2} - A_{i1}) + \frac{A_{i1} + A_{i2}}{2}(\beta_{i2} - \beta_{i1}),$$

which can be verified by expanding the right-hand side and collecting terms. Summing over $i \in S$ yields (15). □

Remark 3.2 (Limitation). Because $\overline{\beta}_i$ and $\overline{A}_i$ embed both periods, each term mixes "drivers" mechanically: the decomposition is symmetric but does not fully isolate whether productivity changes or share changes are the primitive driver.

3.4 FHK decomposition (within, between, cross, entry, exit)

Let A_1 denote aggregate productivity in period 1 (the base reference level). Define $\Delta A_i := A_{i2} - A_{i1}$ and $\Delta\beta_i := \beta_{i2} - \beta_{i1}$ for survivors.

Proposition 3.3 (Canonical FHK decomposition)**.** *Aggregate change decomposes as*

$$\Delta A = \underbrace{\sum_{i\in S}\beta_{i1}\Delta A_i}_{within} + \underbrace{\sum_{i\in S}\Delta\beta_i(A_{i1} - A_1)}_{between} + \underbrace{\sum_{i\in S}\Delta\beta_i\Delta A_i}_{cross} + \underbrace{\sum_{i\in E}\beta_{i2}(A_{i2} - A_1)}_{entry} - \underbrace{\sum_{i\in X}\beta_{i1}(A_{i1} - A_1)}_{exit}. \tag{16}$$

Proof. Start from the identity

$$\Delta A = \left(\sum_{i\in S}\beta_{i2}A_{i2} - \sum_{i\in S}\beta_{i1}A_{i1}\right) + \sum_{i\in E}\beta_{i2}A_{i2} - \sum_{i\in X}\beta_{i1}A_{i1}.$$

For the survivor component, add and subtract $\sum_{i\in S}\beta_{i2}A_{i1}$:

$$\begin{aligned}\sum_{i\in S}\beta_{i2}A_{i2} - \sum_{i\in S}\beta_{i1}A_{i1} &= \sum_{i\in S}\beta_{i2}(A_{i2} - A_{i1}) + \sum_{i\in S}A_{i1}(\beta_{i2} - \beta_{i1}) \\ &= \sum_{i\in S}(\beta_{i1} + \Delta\beta_i)\Delta A_i + \sum_{i\in S}(A_1 + (A_{i1} - A_1))\Delta\beta_i \\ &= \sum_{i\in S}\beta_{i1}\Delta A_i + \sum_{i\in S}\Delta\beta_i\Delta A_i + \sum_{i\in S}\Delta\beta_i(A_{i1} - A_1) + A_1\sum_{i\in S}\Delta\beta_i.\end{aligned}$$

Now incorporate entry/exit using the base reference A_1:

$$\sum_{i\in E}\beta_{i2}A_{i2} = \sum_{i\in E}\beta_{i2}(A_{i2} - A_1) + A_1\sum_{i\in E}\beta_{i2},$$

$$\sum_{i\in X}\beta_{i1}A_{i1} = \sum_{i\in X}\beta_{i1}(A_{i1} - A_1) + A_1\sum_{i\in X}\beta_{i1}.$$

Because total shares sum to one in each period,

$$\sum_{i\in S}\Delta\beta_i + \sum_{i\in E}\beta_{i2} - \sum_{i\in X}\beta_{i1} = \left(\sum_{i\in S}\beta_{i2} + \sum_{i\in E}\beta_{i2}\right) - \left(\sum_{i\in S}\beta_{i1} + \sum_{i\in X}\beta_{i1}\right) = 1 - 1 = 0.$$

Hence all A_1 terms cancel, yielding (16). □

Remark 3.3 (Economic meaning of the cross term). The cross term $\sum_{i\in S}\Delta\beta_i\Delta A_i$ captures the interaction: firms that both increase productivity and gain share (or decrease both) contribute positively; opposite movements contribute negatively.

4 Olley–Pakes (OP) decomposition and allocation efficiency

4.1 Static OP identity

Definition 4.1 (Simple mean and covariance term)**.** Let $N_t := |\mathcal{I}_t|$. Define the simple mean

$$\mu_t := \frac{1}{N_t}\sum_{i\in\mathcal{I}_t}A_{it},$$

and interpret β_{it} as a share weight with $\sum_i \beta_{it} = 1$.

Proposition 4.1 (OP decomposition)**.** *Aggregate productivity* (13) *can be written as*

$$A_t = \mu_t + \mathrm{Cov}\big(\beta_{it}, A_{it}\big), \tag{17}$$

where covariance is computed across firms in period t treating each firm as one observation.

Proof. Let $\bar{\beta}_t := \frac{1}{N_t}\sum_i \beta_{it} = \frac{1}{N_t}$ because $\sum_i \beta_{it} = 1$. Then

$$\begin{aligned}\mathrm{Cov}(\beta_{it}, A_{it}) &:= \frac{1}{N_t}\sum_i (\beta_{it} - \bar{\beta}_t)(A_{it} - \mu_t) \\ &= \frac{1}{N_t}\sum_i \beta_{it}A_{it} - \bar{\beta}_t\mu_t - \mu_t\bar{\beta}_t + \bar{\beta}_t\mu_t \\ &= \frac{1}{N_t}\sum_i \beta_{it}A_{it} - \frac{1}{N_t}\mu_t.\end{aligned}$$

Multiplying both sides by N_t gives

$$N_t\mathrm{Cov}(\beta_{it}, A_{it}) = \sum_i \beta_{it}A_{it} - \mu_t,$$

which rearranges to (17). □

Remark 4.1 (Interpretation)*.* μ_t is "average technology." The covariance term is "allocation efficiency": it is positive when high-productivity firms command above-average shares.

4.2 Dynamic OP with entry and exit: group-based accounting

We now connect OP to the two-period, changing-firm-set setting.

Definition 4.2 (Group shares and within-group shares)**.** In period 1, partition firms into survivors S and exiters X. Define group output shares

$$\omega_{S1} := \sum_{i\in S}\beta_{i1}, \qquad \omega_{X1} := \sum_{i\in X}\beta_{i1}, \qquad \omega_{S1} + \omega_{X1} = 1,$$

and within-group shares

$$\beta_{i|S,1} := \frac{\beta_{i1}}{\omega_{S1}}, \qquad i \in S; \qquad \beta_{i|X,1} := \frac{\beta_{i1}}{\omega_{X1}}, \qquad i \in X.$$

In period 2, partition into survivors S and entrants E with analogous definitions ω_{S2}, ω_{E2} and $\beta_{i|S,2}, \beta_{i|E,2}$.

Then

$$A_1 = \omega_{S1}\sum_{i\in S}\beta_{i|S,1}A_{i1} + \omega_{X1}\sum_{i\in X}\beta_{i|X,1}A_{i1}, \qquad A_2 = \omega_{S2}\sum_{i\in S}\beta_{i|S,2}A_{i2} + \omega_{E2}\sum_{i\in E}\beta_{i|E,2}A_{i2}. \tag{18}$$

Within-survivor OP. Define within-survivor simple means and covariances:

$$\mu_{S1} := \frac{1}{|S|}\sum_{i\in S}A_{i1}, \qquad \mu_{S2} := \frac{1}{|S|}\sum_{i\in S}A_{i2},$$

$$\mathrm{Cov}_{S1} := \mathrm{Cov}\big(\beta_{i|S,1}, A_{i1}\big), \qquad \mathrm{Cov}_{S2} := \mathrm{Cov}\big(\beta_{i|S,2}, A_{i2}\big),$$

and similarly for exiters/entrants if needed. Applying (17) within the survivor group:

$$\sum_{i\in S}\beta_{i|S,t}A_{it} = \mu_{St} + \mathrm{Cov}_{St}, \qquad t \in \{1, 2\}.$$

Substitute into (18):

$$A_1 = \omega_{S1}(\mu_{S1} + \mathrm{Cov}_{S1}) + \omega_{X1}\bar{A}_{X1}, \qquad A_2 = \omega_{S2}(\mu_{S2} + \mathrm{Cov}_{S2}) + \omega_{E2}\bar{A}_{E2}, \tag{19}$$

where $\bar{A}_{X1} := \sum_{i\in X} \beta_{i|X,1} A_{i1}$ and $\bar{A}_{E2} := \sum_{i\in E} \beta_{i|E,2} A_{i2}$ are group-weighted means for exiters/entrants.

Proposition 4.2 (Dynamic OP: a four-part decomposition template)**.** *From* (19)*, one can always write*

$$\Delta A = \underbrace{\omega_{S2}\mu_{S2} - \omega_{S1}\mu_{S1}}_{\textit{survivor mean (technology) change}} + \underbrace{\omega_{S2}\mathrm{Cov}_{S2} - \omega_{S1}\mathrm{Cov}_{S1}}_{\textit{survivor allocation change}} + \underbrace{\omega_{E2}\bar{A}_{E2}}_{\textit{entry contribution}} - \underbrace{\omega_{X1}\bar{A}_{X1}}_{\textit{exit contribution}}. \tag{20}$$

Proof. Subtract A_1 from A_2 using (19) and regroup terms by construction. □

Remark 4.2 (Conceptual limitation). (20) captures within-survivor reallocation (via Cov_{St}) but does not isolate cross-group reallocation caused by creative destruction (changes in the group shares ω_{gt} interacting with group productivity levels).

5 A group-level OP extension (capturing cross-group reallocation)

A natural fix is to treat groups themselves as "units" and apply OP at the group level.

5.1 Group-level OP in each period

In period 1 define two groups $g \in \{S, X\}$ with group productivity $\bar{A}_{g1}$ (e.g., group-weighted mean) and group share ω_{g1}. In period 2 define groups $g \in \{S, E\}$ with $(\bar{A}_{g2}, \omega_{g2})$.

Define the *two-group* simple mean in each period:

$$\mu_1^G := \frac{\bar{A}_{S1} + \bar{A}_{X1}}{2}, \qquad \mu_2^G := \frac{\bar{A}_{S2} + \bar{A}_{E2}}{2}.$$

Then OP at the group level yields

$$A_1 = \mu_1^G + \mathrm{Cov}(\omega_{g1}, \bar{A}_{g1}), \qquad A_2 = \mu_2^G + \mathrm{Cov}(\omega_{g2}, \bar{A}_{g2}), \tag{21}$$

where the covariance is computed across the two groups.

Proposition 5.1 (Five-part decomposition with cross-group allocation)**.** *Combining within-group OP for survivors and group-level OP across groups yields a decomposition of ΔA into:*

1. *survivor within-group mean change,*
2. *survivor within-group covariance change,*
3. *entrant-vs-survivor productivity gap weighted by entrant share,*
4. *survivor-vs-exiter productivity gap weighted by exiter share,*
5. *change in the* between-group *covariance* $\Delta\mathrm{Cov}(\omega_g, \bar{A}_g)$.

Proof (algebraic construction). Start from $\Delta A = (\mu_2^G - \mu_1^G) + \left[\mathrm{Cov}(\omega_{g2}, \bar{A}_{g2}) - \mathrm{Cov}(\omega_{g1}, \bar{A}_{g1})\right]$ using (21). Next, expand each group productivity $\bar{A}_{S1}, \bar{A}_{S2}$ into survivor simple mean plus survivor within covariance via (17) applied within S. Finally, rewrite $\mu_2^G - \mu_1^G$ in terms of differences between survivor and entrant/exiter group means and use $\omega_{S1} + \omega_{X1} = 1$ and $\omega_{S2} + \omega_{E2} = 1$ to collect terms. Each of the five items arises as one collected block. □

Remark 5.1 (Economic reading). The fifth term is the missing "creative destruction reallocation" channel: resources shift between groups (survivors vs entrants/exiters), and the productivity ranking of groups matters.

6 Sectoral (industry) decomposition: APPF logic

Aggregate productivity growth can also be decomposed at the sector level, separating:

- **within-sector TFP growth** (technology or efficiency improvements inside sectors),
- **between-sector reallocation** (labor/capital shifting toward higher-productivity sectors).

A generic accounting identity for aggregate TFP growth (in growth-rate form) is:

$$\Delta \log A^{\mathrm{agg}} = \sum_s \bar{v}_s \, \Delta \log A_s + \underbrace{\text{reallocation terms}}_{\text{changes in sector weights interacting with sector productivity levels}}, \tag{22}$$

where $\bar{v}_s$ is an appropriate average weight (e.g., a Tornqvist weight built from sector value-added shares).

Empirically, an application over a long horizon can find that most aggregate TFP growth is due to within-sector TFP growth, with a non-negligible contribution from reallocation (labor reallocation often larger than capital reallocation when labor mobility reforms are important).

7 Investment allocation and the Wurgler elasticity

A different but related way to measure allocation efficiency is to ask whether investment flows toward industries with better growth opportunities.

7.1 Within-year vs between-year elasticities

Let $I_{j,t}$ be investment in industry j at time t (within a country), and let $VA_{j,t}$ be value added. Define growth rates (for example)

$$g_{j,t}^{I} := \Delta \log I_{j,t}, \qquad g_{j,t}^{VA} := \Delta \log VA_{j,t}.$$

Definition 7.1 (Within-year elasticity)**.** Estimate

$$g_{j,t}^{I} = \alpha_t + \eta^{\mathrm{within}} \, g_{j,t}^{VA} + \varepsilon_{j,t}, \tag{23}$$

where α_t are year fixed effects. The coefficient η^{within} captures whether, *within the same year*, industries with higher value-added growth receive higher investment growth.

Definition 7.2 (Between-year elasticity)**.** Compute industry averages within year t,

$$\bar{g}_t^{I} := \frac{1}{J} \sum_{j=1}^{J} g_{j,t}^{I}, \qquad \bar{g}_t^{VA} := \frac{1}{J} \sum_{j=1}^{J} g_{j,t}^{VA},$$

and regress

$$\bar{g}_t^{I} = a + \eta^{\mathrm{between}} \, \bar{g}_t^{VA} + u_t. \tag{24}$$

The coefficient η^{between} captures whether overall investment growth responds to overall growth opportunities across years.

Remark 7.1 (Connection to misallocation). A low within-year elasticity suggests weak reallocation of capital across industries in response to relative opportunities, echoing the idea that wedges/frictions can decouple factor flows from productivity signals.

8 Integrated checklist (how everything connects)

- **Aggregation with heterogeneity**: Micro technology aggregates through demand, markups, and input allocation; wedges distort allocation.
- **Misallocation statistic**: In many models, dispersion of $\log \mathrm{TFPR}$ is a sufficient summary of within-sector misallocation.
- **Accounting decompositions**: In the data, aggregate productivity changes can be decomposed into within-firm (technology), between-firm (reallocation), cross (interaction), and net entry (creative destruction) terms.
- **OP covariance lens**: The covariance between shares and productivity provides a clean measure of allocation efficiency at a point in time; dynamic OP variants track changes over time and across groups.
- **Sectoral and investment perspectives**: APPF-type decompositions quantify reallocation across sectors; Wurgler elasticities quantify whether investment responds to opportunity signals.

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